

Geometry of Particle Methods

Vanilla mean-field WGF: fix μ_0 , sample $X_0^{(1)}, X_0^{(2)}, \dots, X_0^{(n)} \sim \mu_0$ independently, run the dynamics:

$$\begin{cases} \partial_t X_t^{(1)} = \nabla \delta \mathcal{F}[\hat{\mu}_t](X_t^{(1)}) \\ \partial_t X_t^{(2)} = \nabla \delta \mathcal{F}[\hat{\mu}_t](X_t^{(2)}) \\ \vdots \\ \partial_t X_t^{(n)} = \nabla \delta \mathcal{F}[\hat{\mu}_t](X_t^{(n)}) \end{cases}$$

$$\hat{\mu}_t = \frac{1}{n} \sum_{i=1}^n \delta_{X_t^{(i)}}$$

with $X_{t=0}^{(i)} = X_0^{(i)}$, $i = 1 \dots n$.

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Lambert et al. 2022: introduce “smoothed particles” for VI.

Goal: for $\pi(x) \propto e^{V(x)}$, estimate $\hat{\pi}$,

$$\arg \inf_{\hat{\pi} \in \mathcal{G}_k} \text{KL}(\hat{\pi} \mid \pi) \quad \mathcal{G}_k := \left\{ \frac{1}{k} \sum_{i=1}^k \mathcal{N}(\mu_i, \Sigma_i) : \mu \in \mathbb{R}^d, \Sigma \in S_+^d \right\}$$

“Wasserstein over Wasserstein” structure: take $\mathcal{M} = \mathcal{G}_1$

$\implies \mathcal{G}_k$ are the k -particle empirical msrs. in $\mathcal{P}(\mathcal{M})$!

further explored: Petit-Talamon et al. 2025, Bonet et al. 2025